PROTYAY DEY

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EDUCATION

University at Buffalo — New York, US
PhD in Computer Science and Engineering, CGPA: 3.79 / 4.0 — *Aug 2025 – Present*

Indian Institute of Technology Ropar — Punjab, IN
MTech in Artificial Intelligence, CGPA: 9.09 / 10.0 — *Aug 2022 – Jun 2024*

University of Engineering & Management, Jaipur — Rajasthan, IN
BTech in Computer Science and Engineering, CGPA: 9.27 / 10.0; Gold Medalist — *Aug 2017 – Jun 2021*

RESEARCH EXPERIENCE

Computer Vision and Machine Learning Lab, UB (SUNY) — New York, US
Graduate Research Student – Generative AI, Multi-Agent Systems — *Aug 2025 – Present*

- Developed a **geometry-aware density** and **calibration framework** for singular correlation matrices, deriving closed-form densities with theoretical guarantees and achieving **2.73**× precision, and **8.1**× recall across **9** neuroimaging settings.
- Developed **SCAPE** for decentralized SLM multi-agent reasoning, achieving up to **+16.4 points** over token-budget-matched baselines and consistent gains across **3 model families**.
- Proposed **EAD**, a label-free runtime controller for SLM multi-agent reasoning across **6 benchmarks and 14 baselines**, achieving **+7.40 percentage points overall gain**.
- Conceptualized **MAGNET**, a manifold-aware graph diffusion model for brain connectome generation across **3 cohorts**, improving conditional fidelity by up to **21%**.

Computer Vision and Pattern Recognition Lab, IIT Ropar — Punjab, IN
Junior Research Fellow – Federated Learning, Deep Learning — *Apr 2025 – Jun 2025*

- Architected FedRWA for disjoint-label federated learning, evaluated on **12 clients and 313 classes**.
- Achieved **18.6%** accuracy gain and **57.8%** lower client-wise variance across 3 backbones.

Image Processing, Security & Analytics Labs, IIT Ropar — Remote
Research Intern – Applied Deep Learning — *Jul 2024 – Oct 2024*

- Developed **SPAIR-Swin** using **high-entropy patches** and evaluated across 138 camera models.
- Improved patch-level SCMI accuracy by up to **20.36%** across 4 forensic benchmarks.

mVizn Pte. Ltd. — Remote
Machine Learning Intern – 3D Computer Vision — *Jan 2024 – Mar 2024*

- Increased indoor point-cloud classification by **25%** using PointNet++.
- Improved IoU for underrepresented classes by **30%–45%** through class-imbalance handling.

PUBLICATIONS & MANUSCRIPTS

- **P. Dey**, R. Chakraborty, V.S. Lokhande. "Ambient Densities on Singular Correlation Strata with Applications in Generative Neuroimaging" *Under Review, AISTATS 2027.*
- A.Roy*, **P. Dey***, M. Babbitt, R. Chakraborty, L. Varshney, V.S. Lokhande. "Privacy Considerations in Multi-Institutional Data Harmonization Through a Category-Theoretic Lens." *Under Review, ICLR 2027.*
- **P. Dey***, A. Roy*, H. Cho, W. H. Kim, V.S. Lokhande. "MAGNET: Manifold-Aware Graph Diffusion Network for Connectome Generation." NeurIPS, 2026.
- **P. Dey**, R. Chakraborty, A. S. Jadhav, K. Rana, G. Sharma, P. Goyal, "Camera Model Identification with SPAIR-Swin and Entropy-Based Non-Homogeneous Patches," *Under Review ACM TOMM*, 2026.
- K. Rana*, A. S. Jadhav*, **P. Dey***, V. Rathi, P. Goyal, G. Sharma, "SCMI30-IITRPR: Smartphone Camera Model Identification Dataset Comprising Similar and Random Content Image Sets," CVIP, 2026.
- **P. Dey**, A. S. Jadhav, K. Rana, "An Effective CNN-Based Approach for Synthetic Face Image Detection in Pre-Social and Post-Social Media Context," CVIP, 2023.

TECHNICAL SKILLS

Programming: Python, C++, C, SQL, LaTeX, JavaScript, TypeScript
ML: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV
Research: Small Language Models, Agentic AI, Multi-Agent Systems, LLM Reasoning, Transformers, Diffusion Models, Generative AI, Optimization
Tools: Git, Linux, GitHub, Jupyter, VS Code, Hugging Face, vLLM